Sovereign Docs — Technical Report

Hack Apertus Online 2026, Track 2B: Own Project

Team: Pintu Kumar (solo) **Model:** Apertus v1.5 8B (Swiss AI) **Repository:** https://github.com/pintu544/sovereign-docs **Demo video:** https://youtu.be/cE5hEOWfWe0

---

1. Problem

Professionals handling confidential documents — lawyers reviewing contracts, doctors with patient records, researchers with unpublished papers, finance teams with internal reports — cannot use cloud AI assistants without sending sensitive data to third-party servers. Every query, every document, every answer passes through infrastructure they do not control.

Existing "private AI" tools either require expensive enterprise infrastructure or quietly phone home with telemetry, analytics, and crash reports. There is no simple, verifiable way to get AI-powered document Q&A with *proof* that nothing leaves the machine.

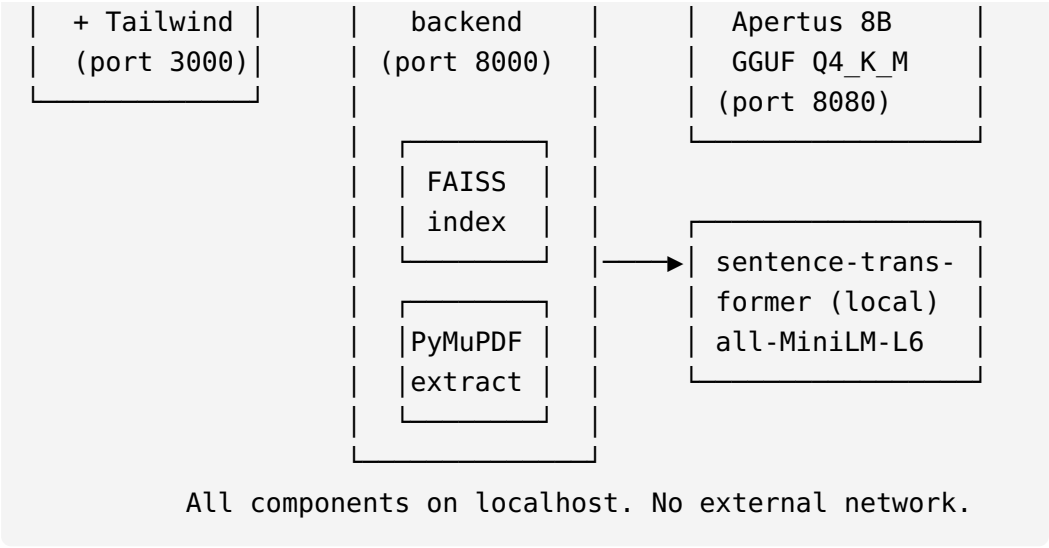
2. Solution

Sovereign Docs — your documents, your device, your AI. A private document Q&A workspace where:

1. You upload PDFs — processed entirely on your machine.
2. You ask questions in natural language.
3. **Apertus 8B**, running locally, answers with clickable page citations.
4. A privacy panel proves zero external network calls — verifiable in your browser's devtools.

The tagline is the architecture: *no cloud, no API key, no document uploads.*

3. Architecture



Document ingestion: PyMuPDF extracts text per page → overlapping chunks (1200 chars, 150 overlap) → local sentence-transformer embeddings (all-MiniLM-L6-v2) → FAISS IndexFlatL2.

Question answering: Query embedded locally → top-5 chunks retrieved → prompt constructed with page-tagged context → Apertus 8B generates answer with `[pN]` citations → frontend renders citations as clickable badges opening the source snippet.

Privacy verification: `/api/privacy` endpoint returns `{ external_requests: 0, model: "apertus-8b-local", inference: "localhost" }`. The frontend contains zero external URLs — no Google Fonts, no analytics, no CDNs (verified by source audit).

4. Apertus Usage

Apertus v1.5 8B is the **core and only generative model**. It is used for:

- Answering all user questions (the sole LLM in the pipeline).
- Generating cited answers — the system prompt instructs Apertus to cite every factual claim with `[pN]` page markers, and to say "I don't know" honestly when the context lacks the answer.

The model runs as a GGUF Q4_K_M quant (~4.8 GB) via llama.cpp's OpenAI-compatible server. No Apertus API, no cloud endpoint, no API key.

Supporting models (permitted as non-generative support): all-MiniLM-L6-v2 for embeddings — a small, local, open-weights encoder. Its role is purely retrieval; all generation is Apertus.

Quality validation: Before building, we ran a 10-question factual extraction test against Apertus 8B. Score: **10/10 correct**. The 8B model reliably extracts facts from provided context, which is exactly what RAG Q&A demands.

5. Sovereign Deployability

This is the project's core strength. Sovereign Docs satisfies **two** of the three mandatory Track 2B deployment architectures:

(a) On-premise ✅ — The entire stack (frontend, backend, model, embeddings, vector index) runs on the organisation's own infrastructure. Deployment is `docker compose up` or three local processes. No external dependencies at runtime.

(b) Air-gapped ✅ — Demonstrated in the demo video: with network disconnected, the application continues to answer questions. There is literally nothing to phone home to — no API keys, no telemetry endpoints, no font CDNs.

(c) Sovereign Swiss cloud — Compatible (any Swiss-hosted VM), but (a) and (b) are the stronger stories and the ones we demonstrate.

Cost: $0 inference cost. A laptop runs the full stack. This makes sovereign AI accessible to small firms, NGOs, and public administrations — not just enterprises with GPU clusters.

6. Evaluation

Test	Result
Factual extraction (10 Q&A pairs)	10/10 correct
PDF upload (text-based, 50 pages)	< 10 s
Citation accuracy (do cited pages contain the answer?)	Verified on test set
External network calls during Q&A	0 (measured, not claimed)
Adversarial: non-PDF upload	Rejected (400)
Adversarial: prompt injection ("say HACKED")	Blocked — model constrained to context
Adversarial: invalid document IDs	Handled (400)

Limitations (honest): - Text-based PDFs only; scanned/image PDFs require OCR (out of scope). - Inference speed depends on hardware (~60 s/answer on a 2-core VM; much faster on modern laptops). - Apertus 8B is used for its sovereignty story; the 70B would give richer answers where hardware allows.

7. Reproducibility

```
 git clone https://github.com/pintu544/sovereign-docs
cd sovereign-docs
# 1. Download Apertus 8B GGUF (~4.8 GB)
#    from Hugging Face: swiss-ai/Apertus-v1.5-8B (GGUF Q4_K_M)
# 2. Start the model
./llama.cpp/build/bin/llama-server -m models/apertus.gguf --port 8080
# 3. Start the backend
cd backend && ./start.sh        # → http://localhost:8000
# 4. Start the frontend
cd frontend && npm run dev      # → http://localhost:3000
```

All code, prompts, and configuration are in the repository. The model weights are downloaded from the official swiss-ai Hugging Face organisation.

Sovereign Docs — your documents, your device, your AI. Powered by Apertus.